

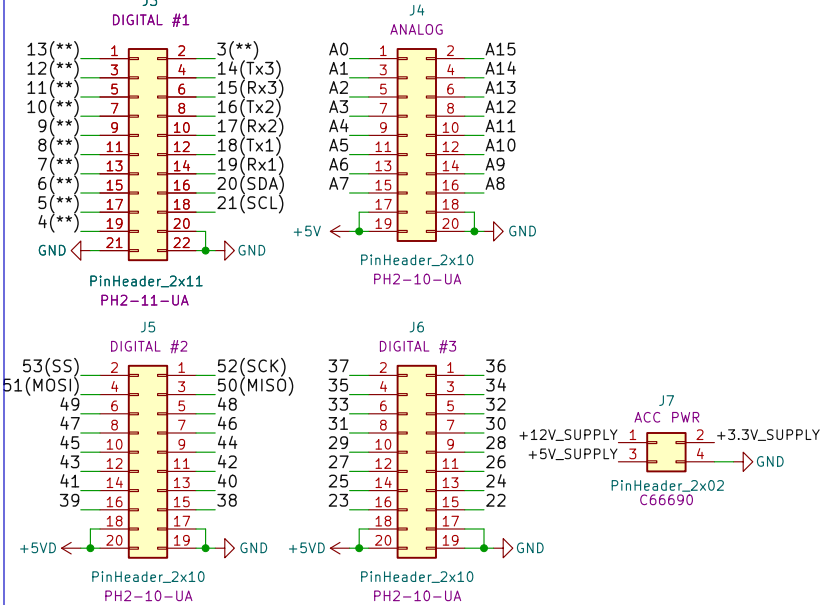
NOTES: (UNLESS OTHERWISE SPECIFIED)

- 1) INSTALL JUMPER (PN: JM-2BK-61 OR SIMILAR) ACROSS JP1 TO ENABLE RS485.
REMOVE JUMPER TO PROGRAM PRO MICRO VIA USB.

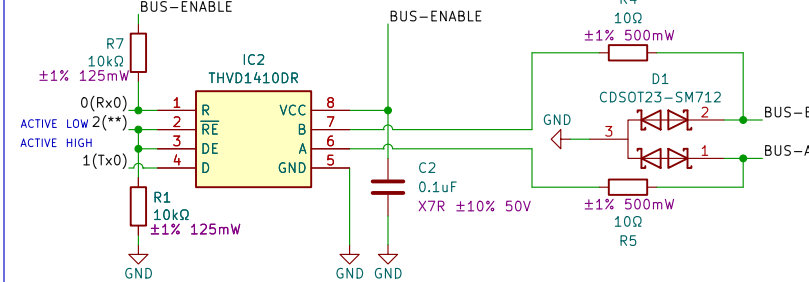
CAUTION: POWER SUPPLY PINS ON BUS OUT CONNECTOR WILL BE ACTIVE REGARDLESS IF DATA IS ENABLED ON THAT CONNECTOR.

- 2) RS485 BUS TERMINATION: DO NOT INSTALL J2 (BUS OUT) CONNECTOR AND ADD A 120OHM THRU-HOLE RESISTOR ACROSS PINS 4&5 OF J2 (BUS OUT) ON LAST SLAVE DEVICE OF RS485 BUS.
- 3) ABSIS RS485 BUS SPECIFICATION
 - 3.1) BUS IS HALF DUPLEX: EITHER TRANSMITTING OR RECIEVING AT ANY GIVEN TIME.
 - 3.2) PIN FUNCTION: R: DATA OUTPUT / D: DATA INPUT / DE HAS INTERNAL PULL-DOWN (DRIVER DISABLED BY DEFAULT) / RE HAS INTERNAL PULL-UP (RECIEVER DISABLED BY DEFAULT)
 - 3.3) CIRCUIT CONTAINS OPEN, SHORT, AND IDLE BUS FAILSAFE WHICH PREVENTS THE SYSTEM FROM INTERPRETING ELECTRICAL NOISE AS VALID DATA.
 - 3.4) THVD1410DR TRANSCEIVER OPERATES WITH TTL LEVEL INPUTS. (LOGIC HIGH: 2V - VCC / LOGIC LOW: 0V - .8VCC)
 - 3.5) CAUTION: AS CABLE LENGTH INCREASES, DATA RATE DECREASES PROPORTIONALLY.

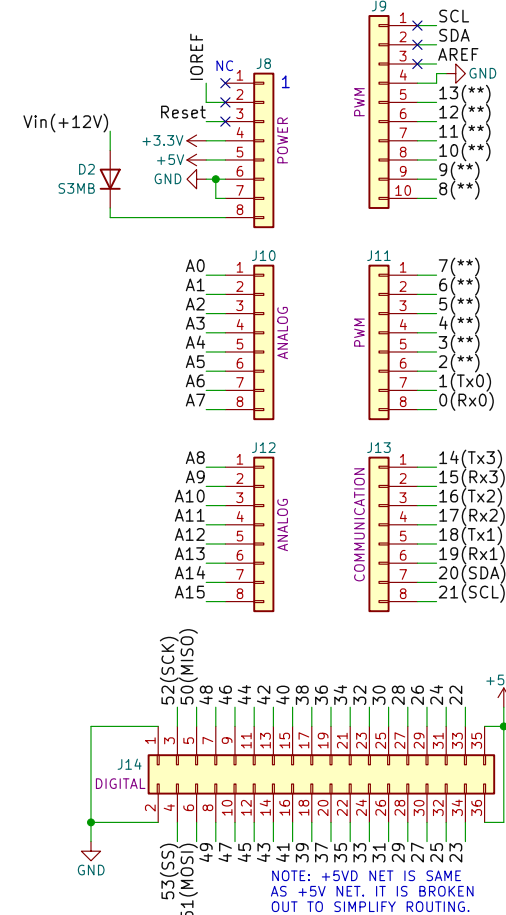
I/O CONNECTORS



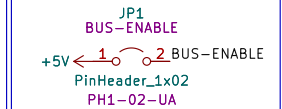
RS485 BUS



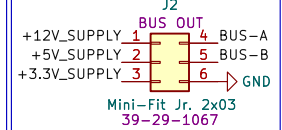
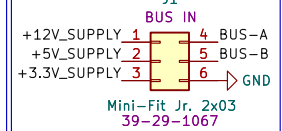
ARDUINO MEGA REV3



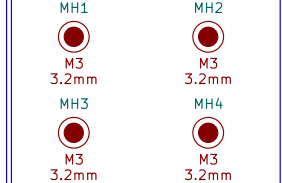
BUS ENABLE



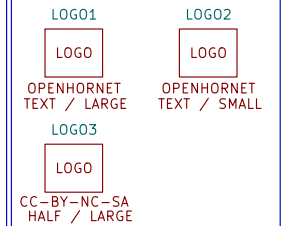
BUS CONNECTORS



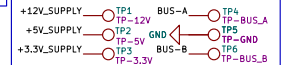
MOUNTING HOLES



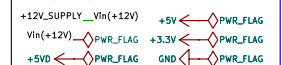
SILKSCREENS



TEST POINTS



POWER FLAGS



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File: ABSIS_Mega.kicad_sch

Title: ABSIS MEGA 2.0

Size: A Date: 2026-06-05

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Rev: 3.0.0

Id: 1/1